

Automatic Docking and Recharging System for Autonomous Security Robot

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Abstract –To keep autonomous mobile robot in continuously working condition power supply is an important issue. Typically, rechargeable batteries may provide few hours of peak usage. Recharging is necessary before the power of the batteries has exhausted. In this paper, the docking station with an auto-recharging device and robot docking mechanism are implemented. We propose a docking control strategy and automatic recharging device for the security robot recharging. We also propose a power prediction algorithm to determine when the robot needs to navigate back to docking station for recharging. An artificial landmark is detected and recognized by the proposed image processing system. Then the geometrical relationship between the robot and the docking station (the depth and orientation) is estimated. The robot will move directly right in front of the docking station. Finally, the robot approaches to the docking station based on the proposed virtual spring model. In this model, we assume that the robot and the docking station are connected by a virtual spring. The compliant forces act both in the direction of the translation deformation and bending. The motion control parameters, which are velocities of the two wheels, can be derived from the model. Experimental results show that the proposed methods successfully perform the docking and automatic recharging.

Index Terms – Security Robot, Vision processing, Docking, Auto-Recharging.

I. INTRODUCTION

With the increasingly research on mobile robot in recently decades, the role of the robot will be more important in the current and future world. Mobile robots are being designed to interact increasingly with human environment, working around humans on a daily basis. The robot systems will be widely employed in many applications, for example, factory automation, dangerous environments, office, hospitals, surgery, entertainment, space exploration, farmland, military, security system, and so on. In this paper, we describe our fully designed and developed mobile robot for security guard indoor [1-4]. With this purpose, the robot should be provided with multi-sensor for the change of the environments in order to take the duty of guarding. Efficient communication between the robot and its outside world is also necessary. In addition, the robot must be capable of long-term autonomy. Energy is of great concern, and without it the robot will become immobilized and useless. How to navigate the robot back to the recharging station and dock into the station are important issues for the long-term functionality of robot. In this paper, we have proposed a virtual spring[5] based docking algorithm, auto-recharging system and docking

mechanism. The power prediction algorithm is also developed.

The first work on robot recharging is made by Grey Walters [6]. They developed the first autonomous recharging mobile robots and employed light to find a hut, which contains a light beacon and a battery recharger. Roufas [7] used four IR LED emitters and one IR receiver to implement the docking. Their method can allow 6 degree of error. In [8], the robot approaches the recharging station using the map and makes contact with specially designed battery recharging system. Some systems are using vision-based method to implement docking [9-12]. The robot designed by Silverman [12] consists of an immobile docking station and the docking mechanism mounted on the robot. Vision and laser ranger-finder are used to find the docking station. The docking station is designed with 2 degree of freedom to tolerate the docking error.

The paper is organized as follows: In Section II, we introduce the system configuration of the security robot, the docking station and robot docking mechanism, auto-recharging device, and the vision system configuration for docking. We present the docking strategy and vision processing in Section III. Section IV presents the experimental results for the docking. Section V presents brief concluding comments.

II. SYSTEM CONFIGURATION

A. Robot System Configuration

The security robot we developed is named as “Chung-Cheng Shin-Kong NO.1”[1-3] and shown in Fig. 1. The robot is specially designed and to be used for the purpose of security patrol and developed by our laboratory. The hardware includes main controller (employed by Pentium-III 933M with 512M RAM), sensory circuits (CCD, wireless LAN, ultrasonic, infrared, body, and power sensors), servomotors, and power system. There are six subsystems in the robot including remote supervisory, obstacle avoidance, motion planning, vision, and sensor systems.

B. Docking Station and Robot Docking Mechanism

The docking station shown in Fig. 2 is a stationary fixture. There are two groups of recharging pins for the robot to dock. They provide different voltages by the auto-recharging device as shown in Fig. 5. The upper one provides 12V/DC with maximum 30A to recharge the batteries for computer and sensory circuits. The lower one provides 36V/DC with maximum 15A to recharge the

batteries for motors and their drivers. Under recharging, the computer is still alive in order to monitor the recharging schedule and depart from the docking station when recharging is completed.

The robot docking mechanism is shown in Fig. 3(a). They are mounted in front of the robot. The recharging adapters are at the inside of the holes. Each hole is shaped as a cone in order to help the docking smoothly. Since both the connection pins and adapters are rigid, the docking station is designed for providing compliance for reasonable robot docking angle and offset. When the robot is approaching to the docking station with offset and docking angle, the guiding stick mounted on the docking station functions to guide the recharging pins for inserting into the adapters. Fig. 4 shows the mechanism of docking station. Extension springs are used to bring the recharging pins back to the center after the robot disengages. A little entry offset and angle are allowed when robot is in the docking process. The maximum rotational angle and horizontal offset are 5 degree and 2 cm respectively.



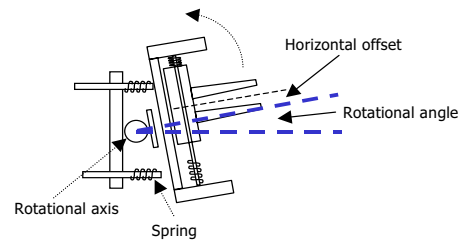
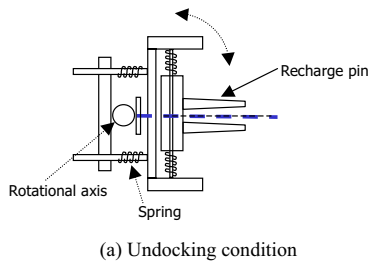
Fig. 1 The security robot "Chung-Cheng Shin-Kong NO.1".



Fig. 2 The docking station



Fig. 3 The robot docking mechanism.



(b) Possible docking condition
Fig. 4 Docking Device with 2 passive DOF.



Fig. 5 The prototype of the auto-recharging device.

C. Auto-Recharging Device

The auto-recharging electrical arrangement schematic is shown in Fig. 6. The recharging flowchart in the docking station is shown in Fig. 7. The robot touches the limit switch that is mounted on docking station. The auto-recharging device can provide the power to the robot after 30 seconds. The current sensor can detect the recharging current value to prevent the faulty status. The current overload represents the condition of electrical system short. The battery saturation represents the batteries to be charged over. No current represents that the charger device has not attached to the security robot. Normal current represents that the security robot is charged with charger device. In the current overload and no current condition, the auto-recharging device must alarm and display faulty status. Then the recharging procedure must be stop. In the battery saturation condition, the recharging status delays 30 seconds. Then the auto-recharging device must cut off recharging current to the security robot.

The hardware block diagram of the auto-recharging device is shown in Fig. 8. The controller is microprocessor, and detects the recharging current using DC type current sensors. The input signals include limit switch and laser source. The output signals contain safety switch, laser light and display. The safety switch may be cut off or switched on the recharging current.

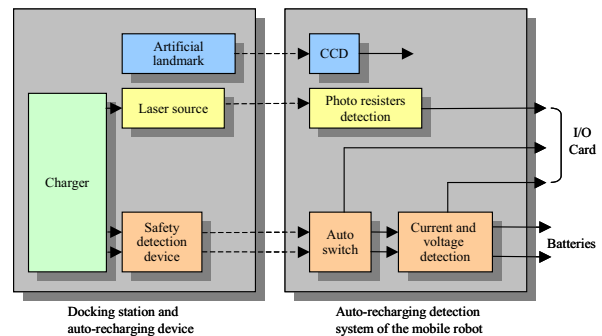


Fig. 6 Schematic of the auto-recharging electrical arrangement schematic.

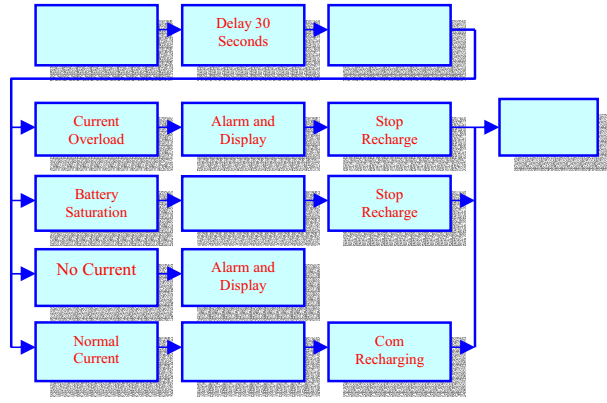


Fig. 7 The flowchart of the control system in auto-recharge process

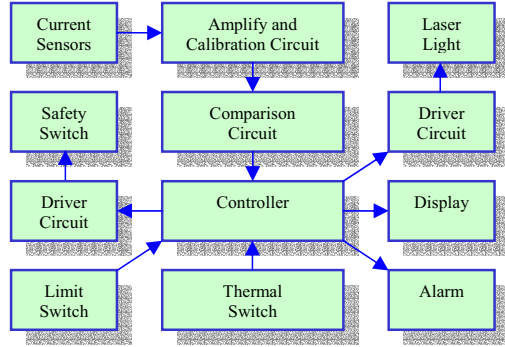


Fig. 8 The hardware block diagram of the auto-recharging device

D. Vision System Configuration

The configuration of vision system for docking is shown in Fig. 9. There is a camera mounted on top of the robot. The video signal is captured by the image frame grabber, which is installed inside the main controller. The acquired image is processed to detect the artificial landmark and locate the site of the docking station. Then the vision system guides the motion to achieve the docking. The motion commands are issued to the motion system through the inter-process communication (IPC). After the motion system executed the motion command, an acknowledge (ACK) command is sent back to the vision system in order to inform the further processing. The vision task is completed while the docking is successful.

In order to estimate the geometrical relationship between the robot and the docking station by inverse perspective projection, the height of the artificial landmark is setup as the same height as the camera, and right on top of the docking station. The height is known in advance. This landmark is specially designed for vision system. It is a navy blue circle. The diameter of this circle is also known.

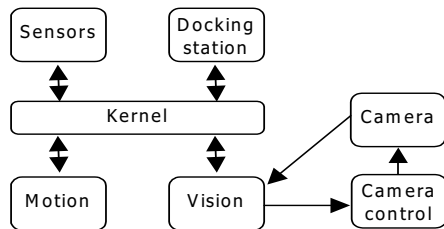


Fig. 9 Relationship between vision and robot system configuration.

III. DOCKING CONTROL

When the power is under the user-defined threshold, the robot should perform the following procedures:

- Step 1. Navigating the robot back to the position near the docking station.
- Step 2. Move the robot to the initial docking position.
- Step 3. Perform the docking by vision system.

The docking strategy is shown in Fig. 10. Robot is navigated to the initially docking position when the docking command is launched. The motion planning system navigates the robot back to the position A. There are three essential steps for vision in guiding the robot to achieve the docking.

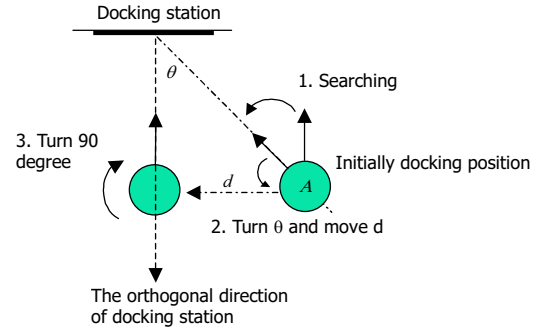


Fig. 10 Docking strategy.

- 1) Vision system searches and locks the artificial landmark.
- 2) Estimate the distance and angle in order to move robot to the front of the docking station.
- 3) Robot approaching control to dock into station by using virtual spring model.

The details are described as follows. First, vision system begins to search the artificial landmark by rotating the robot. The turning direction is predicted by the position estimation according to the internal dead reckoning. While the artificial landmark is detected, vision system issues a stop command to the motion system through IPC. The IPC is implemented by the method of sharing memory.

Second, camera zooming and robot turning are both employed to locate the landmark at the center of the image. The size of the landmark should be zooming as large as possible. It concerns the accuracy of the depth and direction angle from robot to the docking station.

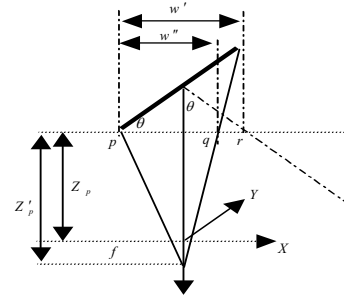


Fig. 11 Geometric relationship.

As we mentioned in the system configuration, we assume that the landmark is placed at the same height as the camera. Also the principle axis of the camera will pass through the central point of the landmark. The direction angle shown in Fig. 10 is estimated by the inverse perspective projection. Refer to Fig. 11, the distance from robot to the docking station can be estimated from the acquired image.

$$\frac{h}{f} = \frac{H}{Z'_p}, Z'_p = \frac{Hf}{h}, Z_p = f - Z'_p, \quad (1)$$

where W and H is the width and height of the landmark respectively. Generally, the distance from robot to the landmark is quite larger than the focus of the camera. Thus we can simplify the distance as follow.

$$Z_p = -Z'_p, \text{ if } Z_p \gg f \quad (2)$$

It is easily to calculate the angle by the width and height of the ellipse. According to the inverse perspective projection, we can obtain the w' (p to r) by the formula below.

$$\frac{H}{h} = \frac{w''}{w}, w'' = \frac{wH}{h}, \quad (3)$$

If the distance of the robot to the docking station is greater than camera focus, then L_1 is almost parallel to the line L_2 . Thus,

$$w'' \cong w', w'' \text{ is the segment from } p \text{ to } r. \\ \cos \theta = \frac{w'}{W} \cong \frac{w''}{W} = \frac{h}{W} = \frac{w}{h}, \quad (4)$$

Here we assume that $W=H$. Thus the angle can be estimated by the width and height of the ellipse.

Finally, the robot approaches to the docking station by adjusting the head direction and velocity of the robot. We propose a virtual spring model to control the robot. This proposed method is derived from the assumption that the docking station and the mobile robot are connected by a virtual spring. Fig. 12 shows the virtual spring model. The origin of the world coordinate is at the docking station. The vector of the mobile robot is represented by follow.

$$x_r = [x_r \quad y_r \quad \theta_r]^T \quad (5)$$

The inputs (d, ϕ) of the virtual spring model are obtained from vision system, where d represents the estimated distance between robot and the docking station and ϕ the angle from the direction of the robot to the virtual spring. Elastic F_1 is proportional to d and acts in the direction of the translational deformation. F_2 is proportional to ϕ and acts in the direction of bending. We assume that the free length of the virtual spring is 0 while the robot is successfully docked. F_1 and F_2 are defined as follows:

$$F_1 = k_1 d \\ F_2 = k_2 \phi \quad (6)$$

Where, k_1 and k_2 are the expansion and bending coefficients of the virtual spring respectively. The dynamic equations are derived as follows:

$$m\dot{v} = -F_1 \cos \phi - F_2 \sin \phi - k_3 v \quad (7)$$

$$I\dot{\omega} = (F_1 \sin \phi + F_2 \cos \phi - k_4 \omega) \frac{L}{i} \quad (8)$$

m is the mass of the robot and I is the moment of inertia of the robot. k_1 and k_2 are the viscous friction coefficients of

translation and rotation respectively. Thus, we obtain follows:

$$\dot{v} = -\frac{F_1}{m} \cos \phi - \frac{F_2}{m} \sin \phi - \frac{k_3}{m} v \quad (9)$$

$$\dot{\omega} = (F_1 \sin \phi + F_2 \cos \phi - k_4 \omega) \frac{L}{i} \quad (10)$$

v_r and v_l represent the velocity of the right and left wheels respectively. w is the distance between the right and left wheels. Based on the $\dot{v}$ and $\dot{\omega}$ from the dynamic equations, v_r and v_l are the control parameters to the robot motion system. They can be obtained from the equation below:

$$\begin{bmatrix} v_r \\ v_l \end{bmatrix} = \begin{bmatrix} 1 & w/2 \\ 1 & -w/2 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad (11)$$

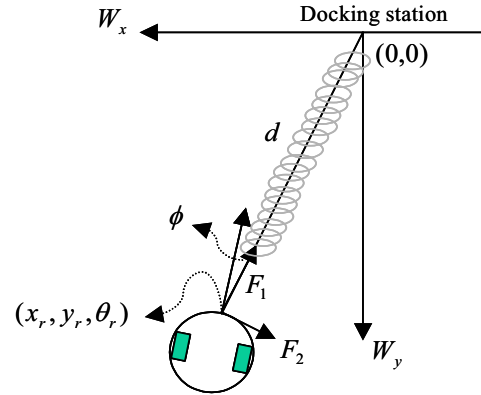


Fig. 12 The virtual spring model.

The inputs (d, ϕ) of the virtual spring model are obtained from vision system as follows:

$$\Delta w = \frac{w/2 - x_c}{w_i} \times w_r \quad (12)$$

$$\phi = \tan^{-1} \left(\frac{\Delta w}{d} \right) \quad (13)$$

Where Δw represents the perpendicular distance from the center of the landmark to the principle line of the camera (as same as the head direction of the robot), w_i is the width of landmark in the image, w_r is the real width of the pattern (pre-known), w is the width of acquired image. d can be obtained by inverse perspective projection.

IV. POWER PREDICTION ALGORITHM

It is important to predict when the battery is necessary to be recharged for long-term autonomy. We develop a power prediction algorithm to determine when the robot needs to navigate back to docking station for recharging. We use the linear regression to fit the curves of discharging and recharging respectively. The linear regression model with distribution of error can be stated as follows:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (14)$$

Where y_i is the value of the response variable in the i th trial. β_0 and β_1 are parameters. x_i is the value of the predictor variable in the i th trial. ε_i is a normally distributed random variable with mean $E\{\varepsilon_i\} = 0$, $i = 1, 2, \dots, n$; and variance $\sigma\{\varepsilon_i\} = \sigma^2$,

$i = 1, 2, \dots, n$. ε_i and ε_j are not only uncorrelated but necessarily independent.

From the equation 14, we know that the response y_i in the i th trial is the sum of two components. One is the constant term $\beta_0 + \beta_1 x_i$ and the other is the random term ε_i . Therefore, y_i is a random variable. Considering simple linear regression model on y versus x is

$$y | x = E(y | x) + \varepsilon | x \quad (15)$$

We assume that

- 1) $E(y | x) = \beta_0 + \beta_1 x$
- 2) The error term $\varepsilon | x$ has mean $E(\varepsilon | x) = 0$, variance $Var(\varepsilon | x) = \sigma^2$ and independent of x .

Therefore, we can rewrite the linear regression model

$$y | x = \beta_0 + \beta_1 x + \varepsilon \quad (16)$$

In the paper, we use the least square estimation to estimate two parameters. First, we express the residual sum of square function to be

$$RSS(h_0, h_1) = \sum_{i=1}^n (y_i - h_0 - h_1 x_i)^2 \quad (17)$$

We want to find h_0 and h_1 to minimize $RSS(h_0, h_1)$.

Thus,

$$RSS(h_0, h_1) = \sum_{i=1}^n (y_i - \bar{y} - (h_0 - \bar{y} + h_1 \bar{x}) - h_1 (x_i - \bar{x}))^2$$

Let $a = h_0 + h_1 \bar{x} \Leftrightarrow h_0 = a - h_1 \bar{x}$

Then we rewrite $RSS(h_0, h_1)$ to $RSS(a, h_1)$

Therefore,

$$\begin{aligned} RSS(h_0, h_1) &= RSS(a, h_1) \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 + n(a - \bar{y})^2 + h_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 - 2h_1 \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) \end{aligned} \quad (18)$$

We define:

$$RXX = \sum_{i=1}^n (x_i - \bar{x})^2 \quad (19)$$

$$RXY = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})$$

$$RYY = \sum_{i=1}^n (y_i - \bar{y})^2$$

Thus, we rewrite (18)

$$RSS(a, h_1) = n(a - \bar{y})^2 + h_1^2 SXX - 2h_1 SXY + SYY \quad (20)$$

According to the definition of least square estimation, we have to minimize $RSS(h_0, h_1)$. Therefore,

$$\min_{a, h_1} RSS(a, h_1) = \min_{a, h_1} (h_1^2 SXX - 2h_1 SXY + SYY), a = \bar{y} \quad (21)$$

We want to find

$$h_1 = \frac{SXY}{SXX} \quad (22)$$

then the minimum value of $RSS(a, h_1)$ is

$$SYY(1 - \frac{SXY^2}{SXX * SYY}) \quad (23)$$

Therefore, we have a theorem here:

The ordinary least square estimate for simple linear regression is

$$\hat{\beta}_1 = \frac{SXY}{SXX}$$

$$\hat{\beta}_2 = \bar{y} - \hat{\beta}_1 \bar{x}$$

(24)

The residual sum of square

$$\begin{aligned} RSS &= \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 * x_i))^2 \\ &= SYY(1 - \gamma^2(x, y)) \\ &= \sum_{i=1}^n \hat{\varepsilon}_i^2 \end{aligned} \quad (25)$$

V. EXPERIMENTAL RESULTS

The docking control algorithm and automatic recharging are implemented in our security robot. We employ a simple and recursive control scheme for recursively testing the docking algorithm and recharging device. The initial position of the robot is docked in the station. The control scheme controls the robot to depart from the station, and moves the robot to the initially docking position. Then the control scheme starts the vision algorithm for docking. After docking and recharging are finished, the control scheme performs the same work again. The successive frames in one trial are shown in Fig. 13. Only the key-point frames are captured from the video.

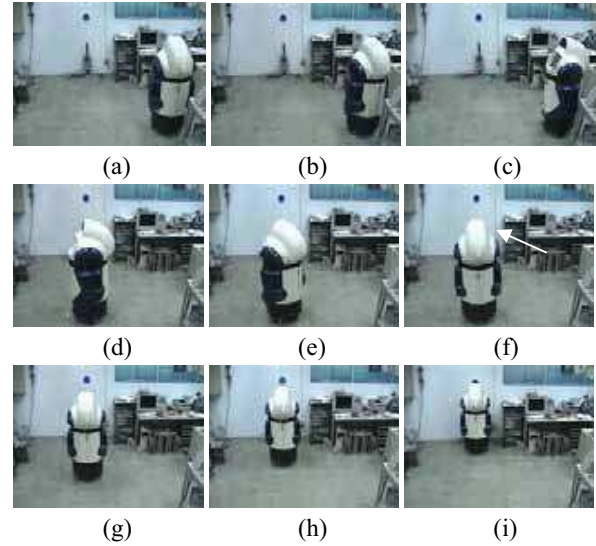


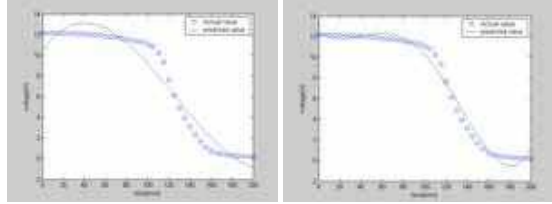
Fig. 13 The sequence of the frames in the figure is left-to-right and top-to-down. (a) The initial docking position. (b) Robot is in the searching state. (c) After the depth and direction of the docking station are estimated, the robot directly moves to the position by rotation and movement (frame c ~ f). Frame g to i exhibit the approaching to the docking station.

To determine the performance of the proposed method, 100 trials were performed. Experimental result shows a 99% success rate for mechanical docking. The successful rate of docking is high. We explain the reason of failure as the quantitation error for region features (a and b in formula 3).

We discharge the battery with a load about 300W (light bulb), and then acquire the related data with five-minute sampling time by the sensor data acquisition system. As the same way, we recharge the battery with no load by battery charger, which can provide 3.5 ampere-hour current (max). Then we can get n set observer data (x_i, y_i) for analysis. The diagram below shows the analysis result. In Case I, we will

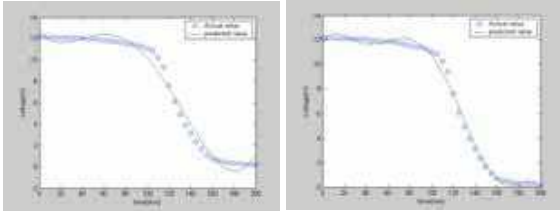
discuss the analysis of discharge, and Fig. 14 shows the experimental results of various orders. In Case II, we will discuss the analysis of recharge. And Fig. 15 shows the experimental results of various orders.

We list these parameter values of the polynomial equation in TABLE I and II. In the experimental results, we can see that the third order polynomial equation is nice for discharging and recharging. In the discharging mode, we can set a threshold voltage 11V. If the battery voltage of the security robot is under 11V, the security robot will navigate itself back to docking station for recharging.



(a). Third Order

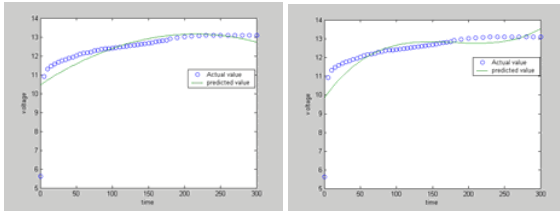
(b). Forth order



(c). Fifth order

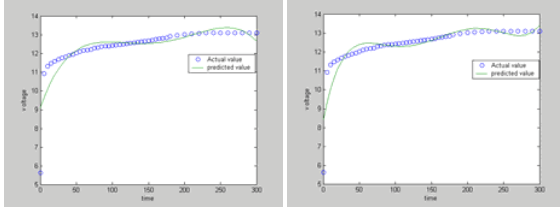
(d). Sixth order

Fig. 14 The polynomial equation of CASE I.



(a). Second Order

(b). Third order



(c). Forth order

(d).Fifth order

Fig. 15 The polynomial equation of CASE II.

TABLE I. Estimate Parameters of Case I

	3	4	5	6
β_0	10.475	12.593	12.768	11.701
β_1	0.1418	-0.0963	-0.1281	0.1675
β_2	-0.0021	0.0034	0.0045	-0.0113
β_3	5.58e-6	-3.76e-5	-5.35e-5	-0.0002
β_4		1.08e-7	1.97e-7	-2.90e-6
β_5			-1.7e-10	1.35e-8
β_6				-2.2 e-11

TABLE II. Estimate Parameters of Case II

	2	3	4	5
β_0	10.471	9.836	9.1233	8.4412
β_1	0.025	0.053	0.106	0.1856
β_2	-5.82e-5	-3e-4	-0.001	-0.003
β_3		5.6e-7	5.06e-6	2.31e-5
β_4			-7.67e-9	-7.71e-8
β_5				9.3e-11

VI. CONCLUSION

Using the docking station with an auto-recharging device and robot docking mechanism developed in our laboratory, the proposed docking control algorithm and virtual spring model are successfully implemented in our security robot. The docking station is designed to accept large docking entry and mount easily. Further research is to investigate multiple security robots cooperating on the security guard and sharing multiple docking stations.

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